

42088 – Industrial Project

Project Plan

Project name:	Automatic, universal push-button actuator, equipped with force gauge
Customer:	Altice Labs
Team members:	<u>Coordinator:</u> Cláudia Almeida, claudiasalmeida@ua.pt , 914102617. <u>Other team members:</u> Rafael Alves, rafael01@ua.pt , 937353627. Gonçalo Peralta, goncaloperalta@ua.pt , 916828627. Pedro Matias, pedrosmatias@ua.pt , 917126548.
Date:	22/10/2024

Revision History

Date	Issue	Description	Author
22/10/2024	Development Costs	Task phasing	Eng. Hélder Alves

Objectives

The purpose of this project is to create a remotely controllable automaton that is capable of pressing buttons, validate their operation, measure the force required to activate them and communicate the results to the server. The solution must ensure that it interacts with various types of buttons (from soft-touch buttons to hard mechanical ones) safely and reliably, while collecting detailed data on the force applied and the button's response.

To achieve the intended end-result, an estimate of the required work was sub-divided into phases. This allows for efficient use of the group's efforts to deliver an end-product with good characteristics and within the defined time frame.

Therefore, this report's primary goal is to give a thorough assessment of how the work will be divided, how the project management will be done, the resources available for its development and the risks involved.

Phase plan

Week	Task		Date	
1	Understand the problem, determine initial requirements and submit proposals with solutions		18/09/2024	Inception
2			25/09/2024	
3	Meeting and Brainstorm with the Company	Team approval of project requirements and initial design concepts	17/10/2024	
4	Definition of the concept to be developed in the project		23/10/2024	
5	Detailed system architecture and list of components finalized		30/10/2024	Elaboration
6	Hardware printed board design		11/06/2024	
7	Test the load cell/resistive sensor to measure the drive force		13/11/2024	Construction
8	Web page development on the Raspberry Pi		20/11/2024	
9	Integrating the actuator and sensor into the mechanical design	Testing basic actuator control using the Raspberry Pi	27/11/2024	
10	Test the API's interaction with the actuator and force feedback. Test universal adjustment on various test devices (routers/gateways)		12/04/2024	
11	Ensure accurate force measurement with load cells		12/11/2024	
12	Testing the API, actuator feedback and force gauge functionality. Debug final problems related to the software or the mechanical fit. Fully functional prototype		25/12/2024	
13			01/01/2025	
15	Collecting final feedback before final construction		01/08/2025	Transition
16				
17	Delivery of the prototype to the client		17/1/2025	

Table 1. Work plan and schedule.

1.1 Objectives of each iteration

As seen in table 1, the project's life cycle was divided into 4 major phases that in turn are divided into iterations. These divisions were pre-defined by the course with basis on an OpenUP framework, with a presentation on the end of each (and in the middle of the "Construction") phase to show all progress achieved.

Inception

This report is written in the end of the Inception phase, and its main objective was to ensure that the entire group had a clear understanding of the proposed work, and the steps required to successfully complete it. More specifically, it was necessary to understand the purpose of the product and the needs

of the devices under test (DUT), in order to create a timeline, a project and team management plan, and to develop a list of necessary components along with the corresponding budget, as well as a list of potential associated risks, ensuring an appropriate response to the problem presented.

Elaboration

In this phase the main objective is to define the system architecture and to transform the high-level vision developed during the Inception Phase into a practical and concrete plan for building the system. This means the system's architecture will be clearly defined, including its structure, components and how they interrelate. This architecture will serve as the blueprint for development, providing a detailed plan of how the system will be built, operated and maintained. This phase focuses on organising the various parts of the system (e.g., actuator, force sensor, Raspberry Pi, API and software interface) in a way that illustrates how they will interact. By defining these interrelationships early, the team can ensure that all components work together seamlessly, and potential integration issues are addressed upfront. Here, proof of concepts and prototypes will play a significant role in mitigating risks. By testing core components (such as the actuator's control and the force sensor's accuracy), potential issues can be identified early, and solutions can be refined before full-scale construction begins.

We will also take a deeper plunge into drawing up the necessary technical specifications for each component. This includes selecting the most suitable technologies (e.g. linear actuators, resistive force sensors), outlining the printed circuit board (PCB), defining the data flow and specifying the signal conditioning circuits and other essential infrastructures.

Since the goal is to take the general vision from Inception and break it down into detailed, actionable plans, technical diagrams, flowcharts, and interface definitions will be created to explain "how" the system will be implemented—whether it's the physical setup of the actuator, the software architecture for the API, or the integration of the force sensor and data transmission protocols.

Construction

The goal of this phase is to start building the product that was designed in the earlier stages. Having the components required in hand, work force is focused on constructing the modules and testing them. The effort required to complete this task is likely high, and as such requires parallelism, minimizing stalls and delivery times.

For this, the development is split into two concurrent development trees, the API and the hardware. Both start with lower-level component development and testing to guarantee a solid foundation - for example the force sensor, and the base operating system for the Raspberry Pi - and rise to the model expected – e.g. proper motor control and working API.

The evolution of each tree will not be done top-down simply in one version, but rather it will be done in iterations that allow the communication between the API and the mechanical interface to be achieved as quickly as possible so to increment functionality. Afterwards, they will be fully tested in conjunction to achieve a working product.

Transition

In this phase, the final product will be delivered and presented, and so it should be a fully functioning product to be tested by users. From there feedback will be used to improve the experience and ensure it works as intended. We will conduct an evaluation of the satisfaction level from our side, as well as from the company and the professors.

2 Budget

This is a raw estimate of the cost associated with this product, but it will be more accurately updated in the elaboration phase.

ID	Category	Description	Value (€)
1	Components	Linear Actuator https://mauser.pt/catalog/product_info.php?products_id=095-0732	32.65
2	Components	Raspberry Pi https://mauser.pt/catalog/product_info.php?products_id=095-3807	19.99
3	Components	Force-Sensing Resistor https://mauser.pt/catalog/product_info.php?products_id=096-6326	14.70
4	Components	8 bit Analog-to-Digital Converter https://mauser.pt/catalog/product_info.php?products_id=001-2384	5.92
Total:			73.26

Table 2. Components and respective costs.

ID	Category	Description	Value (€)
1	Development	Perforated circuit board https://mauser.pt/catalog/product_info.php?products_id=096-4828	5.24
Total:			5.24

Table 3. Development costs.

Altice Labs offered us the opportunity to use their 3D printer and materials, as well as supply the PCB that will be designed, allowing us to focus on development without concerns about those costs. However, it will always be kept in mind that the lower the implementation cost, the easier it will be to replicate the final product.

3 Resources

Our team consists of four students from Electronics and Telecommunications Engineering course, supported by one supervisor from the *Department of Electronics, Telecommunications, and Informatics* (DETI), and two engineers from Altice Labs.

To maximize each member's abilities and to guarantee that everyone is actively involved in the various phases of the project, the group will be split up into pairs or, in certain cases, some members will work alone. It is also important to mention that the group has a project coordinator, who is primarily responsible for managing the team and serves as the liaison between the group and the different supervisors.

We will implement the Scrum project management methodology using the software [Trello](#), with sprints lasting three weeks, shown in Table 2, along with daily short meetings and a more extended weekly meeting with the project supervisor for planning and evaluating the work done. At the end of each sprint, we will have a demo product, where we will discuss for improvements to prepare the next sprint.

Sprints	Weeks	Tasks	Claúdia	Rafael	Pedro	Gonçalo
1	3	Detailed system architecture and list of components finalized	x	x	x	x
		Hardware printed board design		x	x	
		Test the load cell/resistive sensor to measure the drive force	x			x
2	3	Web page development on the Raspberry Pi				x
		Integrating the actuator and sensor into the mechanical design	x	x	x	x
		Testing basic actuator control using the Raspberry Pi				
3	3	Test the API's interaction with the actuator and force feedback. Test universal adjustment on various test devices (routers/gateways)	x	x	x	x
		Ensure accurate force measurement with load cells			x	x
		Testing the API, actuator feedback and force gauge functionality. Debug final problems related to the software or the mechanical fit. Fully functional prototype	x	x	x	x
4	3	Collecting final feedback before final construction	x	x	x	x
		Delivery of the prototype to the client	x	x	x	x

Table 2. Work sprints plan and schedule.

It is worth mentioning that even though sometimes in these sprints plan not all the members are included in a task, this means they will be working in other tasks, not specified, that must be done continually during the semester, like the development of the fixture piece and report's/presentation preparation.

4 Risks and Constrains

Product Risks:

- Linear actuator pushing the buttons too hard, damaging the equipment;
Solution: defining a fixed maximum value of pressure, by limiting the output of the amplifier.
- Poor placement of the automaton on the DUT, creating a misalignment between the actuator and the buttons;
Solution: use a flexible clamp to ease the placement according to the button position.
- Improper placement of the force-sensitive resistor, causing the applied force to not be measured correctly;
Solution: ensure the FSR placement gives an accurate measurement on the given test devices, by testing different locations for its application.

Project Risks:

- PCB taking up to two or three weeks to build, and it may end up not working;
Solution: Use a perforated circuit board instead which will take just a few days to set up.
- Delay on the delivery of the components, slowing the overall progress of the project;
Solution: Approach the problem with temporary equipment.
- Poor team and time management, which can lead to task accumulation;
Solution: Follow the Scrum methodology, helping us to overcome this problem.